



Lab 4. Convolutional Neural Network (CNN)

0. Dataset

Use the Intel Image Classification dataset.

<https://www.kaggle.com/datasets/puneet6060/intel-image-classification>

The dataset contains natural scene images belonging to six classes:

- buildings
- forest
- glacier
- mountain
- sea
- street

Use the seg_train and seg_test folders. The seg_pred folder is not required.

Use the seg_train dataset for training. Split it into training (80%) and validation (20%) sets.

After training the model, evaluate its performance using the seg_test dataset.

1. Dataset Preparation

Prepare the dataset, including:

1. Load the dataset.
2. Resize images to 150×150 if they are not already 150×150 .
3. Normalize pixel values to $[0, 1]$.
4. Split the seg_train dataset into training and validation datasets.

Display several example images from the dataset and report:

- Number of training samples
- Number of validation samples
- Number of test samples
- Number of classes

2. Build a CNN Model

Build a Convolutional Neural Network. The network must include at least:

- 1 Conv2D layer
- 1 MaxPooling layer
- 1 Flatten layer
- 1 Dense layer

Train the model. Report the test accuracy on the seg_test dataset.

3. Modify the CNN Architecture

Modify the CNN architecture. For example, you may:

- Add additional Conv2D layers
- Add additional MaxPooling layers
- Change the number of filters

Train the model. Report the test accuracy on the seg_test dataset.

4. Data Augmentation

Apply data augmentation to the training dataset only. For example:

- horizontal flip
- rotation
- zoom
- brightness adjustment

Train the model. Report the test accuracy on the seg_test dataset.

5. Regularization

Apply regularization techniques such as Dropout layers or EarlyStopping during training. Train the model. Report the test accuracy on the seg_test dataset.

6. Report

Create a short report (1 page) including:

1. Model architecture diagrams
2. Training settings (e.g., epochs, optimizer, loss function)
3. Data augmentation methods used
4. Regularization methods used
5. Compare the test accuracy of the models in Tasks 2–5.

Submission

Submit your source code and report to Canvas and demonstrate your work to the tutor during the lab session.